

Project 1

Every student should submit the code (`studentID_name.zip`), report (`studentID_Name.pdf`), and ppt (`studentID_name.ppt`) on Blackboard system before 23:59 April 28. Report can be written either in English or Chinese.

Q1. OpenClaw for Clinical Data Analysis

Diabetes is one of the most prevalent chronic diseases in the world, affecting millions of individuals and imposing a substantial economic burden. In this project, you will use openClaw to analyze the [Diabetes Health Indicators Dataset](#). The dataset is also provided on the BB system. This file contains the target variable `Diabetes_012` and health indicators.

ALL tasks in Q1 are expected to be completed with OpenClaw. You are also required to **check the correctness** of the OpenClaw output, re-interact with OpenClaw until it generates correct answers.

(1) [2 points] Install [OpenClaw](#) or one of its derivatives (e.g., [AutoClaw](#) which offers a free quota) on your local machine or a cloud server. You can try to install with docker if you like. In your report, briefly describe the overall architecture and working mechanism of the system you use, and introduce 3 skills that are related to data analysis.

(2) [2 point] Create a Spark dataframe from `diabetes_012_health_indicators_BRFSS2015.csv` and check its schema. Report the class distribution of `Diabetes_012`, and perform a descriptive analysis of at least 5 health indicators(`HighBP`, `HighChol`, `BMI`, `PhysActivity`, `Age`). Use Spark SQL API and Spark dataframe API separately, compare the results across different diabetes classes, and summarize which factors appear to be most related to diabetes risk.

(3) [3 points] What observations or conclusions can you draw from this dataset? List three observations or conclusions, with the corresponding codes, tables and graphs for data analyses.

(4) [1 point] Failure case analysis. Record one failure case when using OpenClaw in this question. For the case, describe the task objective, your expected result, the actual incorrect or unsatisfactory result, and your analysis of the possible reasons for failure. Finally, summarize 1 aspect where you think OpenClaw is most valuable for big data analysis tasks.

Q2. Survival Analysis

Survival Analysis is a statistical approach used to analyze the time until an event of interest occurs. This event is often referred to as a "failure" or "event", and it can represent various outcomes depending on the context, such as death, disease recurrence, equipment failure, or any other event that can be timed.

Refer to [this github link](#) or [the databricks link1](#), [link2](#) for the official Spark casebook for survival analysis.

- (1) [3 points] Follow the tutorial step by step on your server. Rewrite all the pyspark code in your own words within one ipynb file. Add enough comments for each code block to explain what the code is for in this block.
- (2) [2 points] Write a report to explain the process of survival analysis in this case. Record all the results in the report.
- (3) [3 points] Create a MySQL database and import the data from the github link mentioned above or from [other databricks industry solution cases](#). Try different questions on these data and prompt the LLMs to generate the corresponding SQL code.

Please list **three** cases that the LLMs fail to generate correct SQL code based on your question, including the database schema, user question, LLM-generated wrong SQL code, and your correct SQL code. Also explain your assumptions that when the LLMs might fail in performing data analysis using SQL.

(4) [3 points] Apply for a free personal domain name through the `Namecheap` or `name.com` from github education pack [here](#). Create a personal website using [github pages](#) and deploy your custom domain on this site. You can read references [here](#) and [here](#), or directly chat with LLMs to generate the required code to build a website. Write a blog to include your data analysis report in task (2).

Note:

1. You can use LLMs to complete these tasks and debug your codes. If so, list all the prompts you used in the report.
2. For task (3), select one LLM from the list of the [Doubao LLM](#) or [Qwen LLM](#) or [OpenAI GPT](#). Following is a prompt example for querying the LLM in the text-to-SQL task. You can also use other reasonable prompts.


```

prompt = ""
You are a smart agent responsible for generating the correct SQL statements based on the following information:
- Database structure information: including table names, fields, relationships between tables (such as foreign keys, etc.).
- User questions: natural language queries or questions.

Your main tasks are:
1. Analyze database structure information:
  - Based on the database structure information, understand the fields and relationships of the table, and build the basic framework of the SQL statement.
2. Generate SQL statements:
  - Based on user questions, query requirements and conditions, tables involved, and auxiliary query conditions, construct complete SQL statements.
3. Verification and optimization:
  - Check whether the generated SQL statement is logical and optimize it if necessary.

### Input:
- Database structure information: including table names, fields, relationships between tables (such as foreign keys, etc.).
- User questions: natural language queries or questions.

### Output:
- Return the result in json format, the format is {"sql": "SQL statement that meets the user's question requirements"}

### Operation steps:
1. Analyze database structure information: build the basic framework of the SQL statement.
2. Generate SQL statements: construct complete SQL statements.
3. Verification and optimization: ensure the logical correctness of the SQL statement and optimize it.

### Note:
- Ensure that the SQL statement accurately reflects the query requirements in the user questions.
- Reasonably construct query logic based on database structure.
- When generating SQL statements, consider all the information provided to ensure the correctness and efficiency of the statements.
- If the user question involves complex query requirements, please consider all requirements to generate SQL statements.

### given the following database schema :
CREATE TABLE continents(
  ContId int primary key ,
  Continent text ,
  foreign key (ContId) references countries (Continent)
);

CREATE TABLE countries(
  CountryId int primary key ,
  CountryName text ,
  Continent int ,
  foreign key (Continent) references continents (ContId)
);

Answer the following question: How many continents are there ?
""

```

Presentation

[6 points] Sampled students will present their work and thoughts within 10 minutes in class. The rest students will record a 10-minutes video for this presentation.